Yash Kumar Singh

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Work Experience

MezZingo — Jan 2026 - Present
Backend Developer Intern — Remote (Bengaluru, India)

- Designed and deployed scalable **Cloud Run backend services** using **Python and Flask** for production systems.
- Developed RESTful APIs for subscription workflows and automated bulk order generation pipelines using **Firestore**.
- Optimized **Firestore queries, batch writes, and transactions** for high-volume workloads improving system performance.
- Implemented structured **error handling, logging, and failure recovery pipelines** improving backend reliability.

Maity Innovations Pvt. Ltd. — Jan 2026
Software Developer Intern (Backend) — Onsite (Durgapur, India)

- Collaborated closely with senior developers on production-level **e-commerce** backend systems in agile development cycles.
- Engineered scalable cart and coupon **engine handling 10+ validation rules** and strict inventory consistency checks.
- Implemented secure **MongoDB transactions** to ensure atomic coupon redemption, eliminating duplicate usage issues.
- Developed and thoroughly tested 20+ **RESTful APIs** and resolved critical auth bugs, **reducing** 500 errors by $\sim 40\%$.

Techxica Technology Pvt. Ltd. — Nov 2025 - Jan 2026
Full Stack Developer Intern — Remote (Maharashtra, India)

- Built scalable product management system managing **40+ products** with admin dashboard and **Razorpay** integration.
- Redesigned company website architecture improving responsiveness and **reducing average page load time by** $\sim 25\%$.
- Resolved critical **payment gateway** issues ensuring **reliable transactions** and 100% successful **webhook** verification.
- Optimized complex **database queries** and **API responses**, improving overall application performance by $\sim 20\%$.

Projects

OpsTrack API | *Node.js, TypeScript, Express.js, MongoDB, Redis, BullMQ, Prometheus, Docker* – GitHub — Mar 2026

- Developed scalable backend supporting authenticated task workflows with **role-based access control (RBAC)**.
- Improved API performance using **Redis caching** and **indexed queries**, minimizing repeated database operations.
- Implemented asynchronous **background jobs** and **rate limiting** to enhance scalability and system reliability.
- Instrumented monitoring exposing $\sim$ 2**ms API latency**, $\sim$ 86–127**ms query times**, and runtime metrics.

BudgetTracker AI | *Next.js 14(TypeScript), MongoDB, Clerk Auth, OpenRouter API* – GitHub | Live Demo — Oct 2025

- Built **LLM-powered** expense categorization across 6+ categories, **reducing manual entry by** $\sim 90\%$.
- Implemented **secure authentication**, protected APIs, and **type-safe backend endpoints** using Clerk.
- Optimized MongoDB access and delivered **analytics dashboards** (bar/pie) for personalized spending insights.

ExpenseSplitter App | *React.js, Node.js, TypeScript, Express.js, MongoDB, Socket.io* – GitHub | Live Demo — Sep 2025

- Developed **real-time expense settlement platform** supporting 30+ concurrent groups with $<$ 100ms latency.
- Built secure REST APIs with **JWT authentication** and implemented flexible expense-splitting logic.
- Optimized client-side routing with **lazy loading**, reducing initial bundle size and load time by $\sim 30\%$.

Technical Skills

- **Programming Languages:** JavaScript (ES6+), TypeScript, C, C++, Python
- **Web Technologies:** React.js, Next.js, Tailwind CSS, Node.js, Express.js, Flask, WebSockets, Socket.io, BullMQ
- **Databases & Authentication:** MongoDB, MySQL, Redis, Firestore, JWT, NextAuth.js, Clerk
- **DevOps & Cloud:** Docker, GitHub Actions (CI/CD), Prometheus, Grafana, Linux, Google Cloud Platform (GCP), Vercel
- **Tools & Version Control:** Git, GitHub, Postman, Hoppscotch

Achievements

Problem Solving	Solved **200+ LeetCode problems** across arrays, strings, linked lists, trees, graphs, recursion, and dynamic programming.
LeetCode Consistency	Earned **100+ Days LeetCode Badge** through consistent daily problem solving
Competitive Programming	Achieved **1300+ CodeChef rating** through regular competitive programming practice
Smart India Hackathon	Participated in **Smart India Hackathon 2025**, building disaster response drone dashboard with GPS integration

Education

National Institute of Technology Durgapur, India — Aug 2024 - May 2028
Bachelors of Technology in Chemical Engineering — CGPA: 7.5/10.0

Relevant Coursework

Core CS: Computer Programming, Data Structures and Algorithms, Object Oriented Programming, Database Management Systems, Operating Systems

Position of Responsibility

Student Alumni Interaction Cell (SAI Cell), NIT Durgapur — Feb 2025 - May 2025
Member

- Organised **Alumination 2025** and coordinated alumni networking sessions, cross-team planning, and on-site logistics.
- Guided **500+ students** during alumni sessions, ensuring structured coordination and effective overall engagement.